

REVISION 2026

# Construction Materials Lab

Diploma in Civil Engineering

Code: 2018

Credits: 1.5

Semester: 2

## Course Outcomes (CO)

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- **CO1:** Identify the physical properties of bricks and prepare cement mortar. (Apply)
- **CO2:** Identify soil types and flooring tiles. (Apply)
- **CO3:** Identify special construction materials and determine bulking of sand. (Apply)

## Module 1: Bricks & Mortar

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### Expt 1: Dimension & Density of Bricks

Measure dimensions of 10 bricks and find average dimension, weight, and density.

### Expt 2: Field Test on Bricks

Perform field tests including dropping, striking, and scratching by nail.

## Expt 4: Setting out of Building

Setting out a small building in the field using ropes and pegs.

### SVG: Brick Dimension Check



Standard Brick (19x9x9 cm)

(Setting out)

## Module 2: Soil & Flooring

### Expt 5: Soil Report

Identify various layers and types of soil in the foundation pit and prepare a report.

### Expt 6: Flooring Tiles

Identify different types of flooring tiles such as vitrified, ceramic, glazed, and mosaic tiles.

## Module 3: Special Materials & Sand

### Expt 8: Bulking of Sand

Determine the increase in volume of sand due to moisture content (Bulking).

(Bulking)

## Practice Model Question Paper

*Practice Model Paper — not an official REV2026 paper*

1. Find the average weight and density of the given 5 bricks.
2. Prepare a cement mortar of 1:4 proportion for the given quantity of sand.

## Source Declaration

**Curriculum Scheme:** Revision 2026

**Subject Name:** Construction Materials Lab

**Subject Code:** 2018

**Official Source URL:** [SITTTR Link](#)

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